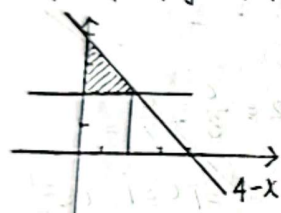


$$\begin{aligned}
 3. (1) F(x, y) &= \int_2^4 \int_0^{6-x-y} k(6-x-y) dx \\
 &= \int_2^4 [k(6x - yx - \frac{1}{2}x^2)]_0^{6-x-y} dy \\
 &= \int_2^4 k(12 - 2y - 2) dy \\
 &= [k(10y - y^2)]_2^4 = k[(40-16) - (20-4)] = 8k = 1 \quad k = \frac{1}{8}
 \end{aligned}$$

$$\begin{aligned}
 (2) P\{X < 1, Y < 3\} &= F(1, 3) = \int_2^3 dy \int_0^1 f(x, y) dx \\
 &= \int_2^3 dy \int_0^1 \frac{1}{8}(6-x-y) dx \\
 &= \int_2^3 dy [\frac{1}{8}(6x - yx - \frac{1}{2}x^2)]_0^1 \\
 &= \int_2^3 \frac{1}{8}(6 - y - \frac{1}{2}) dy = \int_2^3 \frac{1}{8}(\frac{11}{2} - y) dy = (\frac{11}{16}y - \frac{1}{16}y^2)_2^3 \\
 &= \frac{33}{16} - \frac{9}{16} - (\frac{22}{16} - \frac{4}{16}) = \frac{11}{16} - \frac{5}{16} = \frac{6}{16} = \frac{3}{8}
 \end{aligned}$$

$$\begin{aligned}
 (3) P\{X < 1.5\} &= F_X(1.5) = \int_2^4 \int_0^{6-x-y} f(x, y) dy dx \\
 &= \int_2^4 f(x, y) dy = \int_2^4 [\frac{1}{8}(6x - yx - \frac{1}{2}y^2)]_0^{6-x-y} dx \\
 &= \int_2^4 \frac{1}{8}(6x - x^2 - \frac{1}{2}x^2) dx = \int_2^4 \frac{1}{8}(6x - \frac{3}{2}x^2) dx \\
 &= \frac{1}{8}[(24 - 4x^2) - (12 - 2x^2)]_2^4 = \frac{1}{8}[(24 - 16) - (12 - 4)] = \frac{1}{8}(8 - 8) = 0
 \end{aligned}$$

$$\begin{aligned}
 (4) P\{X+Y \leq 4\} &= P\{Y \leq 4-X\} = \int_0^{4-X} \int_2^4 \frac{1}{8}(6-x-y) dy dx \\
 &= \int_0^2 dx \int_2^{4-x} \frac{1}{8}(6-x-y) dy \\
 &= \int_0^2 dx [\frac{1}{8}(6y - xy - \frac{1}{2}y^2)]_2^{4-x} \\
 &= \int_0^2 dx [\frac{1}{8}(6(4-x) - x(4-x) - \frac{1}{2}(4-x)^2) - \frac{1}{8}(12 - 2x - 2)] \\
 &= \int_0^2 dx [\frac{1}{8}(24 - 6x - 4x + x^2 - \frac{1}{2}(16 - 8x + x^2) - 12 + 2x + 2)] \\
 &= \int_0^2 dx [\frac{1}{8}(24 - 6x - 4x + x^2 - 8 + 4x - \frac{1}{2}x^2 - 10 + 2x + 2)] \\
 &= \int_0^2 dx [\frac{1}{8}(6 - 4x + \frac{1}{2}x^2)] \\
 &= \frac{1}{8} \int_0^2 (6 - 4x + \frac{1}{2}x^2) dx = \frac{1}{8} [6x - 2x^2 + \frac{1}{6}x^3]_0^2 = \frac{1}{8} (12 - 8 + \frac{8}{6}) = \frac{1}{8} (\frac{28}{3}) = \frac{7}{6}
 \end{aligned}$$



$$\int_2^4 dy \int_0^{4-y} (6-x-y) dx$$

$$\begin{aligned}
 &= \int_2^4 dy [6x - \frac{1}{2}x^2 - yx]_0^{4-y} \\
 &= \int_2^4 dy [(4-y)(6 - \frac{1}{2}(4-y) - y)] \\
 &= \int_2^4 dy (4-y)(4 - \frac{1}{2}y) \\
 &= \int_2^4 (16 - 2y - 4y + \frac{1}{2}y^2) dy \\
 &= \int_2^4 (16y - 3y^2 + \frac{1}{6}y^3) dy \\
 &= \frac{1}{8} [64 - 48 + 64 \times \frac{1}{6} - (32 - 12 + \frac{4}{3})] \\
 &= \frac{1}{8} (16 + \frac{32}{3} - 20 - \frac{4}{3}) \\
 &= \frac{1}{8} (\frac{28}{3} - \frac{12}{3}) = \frac{1}{8} \times \frac{16}{3} = \frac{2}{3}
 \end{aligned}$$



$$5. F(x, y) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-x-y}, & x > 0, y > 0 \\ 0, & \text{其他} \end{cases}$$

$$F_X(x) = F(x, \infty) = 1 - e^{-x}, x > 0$$

$$F_Y(y) = F(\infty, y) = 1 - e^{-y}, y > 0$$

$$F_X(x) = \begin{cases} 1 - e^{-x}, & x > 0 \\ 0, & \text{其他} \end{cases}$$

$$F_Y(y) = \begin{cases} 1 - e^{-y}, & y > 0 \\ 0, & \text{其他} \end{cases}$$

$$7. f(x, y) = \begin{cases} 4.8y(2-x), & 0 \leq x \leq 1, 0 \leq y \leq x \\ 0, & \text{其他} \end{cases}$$

$x \geq 0$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$= \int_0^x 4.8y(2-x) dy = [2.4(2-x)y^2]_0^x = 2.4(2-x)x^2$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_y^1 4.8y(2-x) dx$$

$$= 4.8y(2x - \frac{1}{2}x^2) \Big|_y^1 = 4.8y(\frac{3}{2} - 2y + \frac{1}{2}y^2)$$

$$= 2.4y(3 - 4y + y^2)$$

$$9. f(x, y) = \begin{cases} cx^2y, & x^2 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

$$(1) F(x, y) = \int_{-\infty}^y \int_{-\infty}^x cx^2y dy dx = \int_{-1}^1 [cx^2 \frac{1}{2}y^2]_{x^2}^y dx$$

$$= \int_{-1}^1 \frac{1}{2}cx^2(1 - x^4) dx$$

$$= (\frac{c}{6}x^3 - \frac{c}{14}x^7) \Big|_{-1}^1 = (\frac{c}{6} - \frac{c}{14}) \times 2 = \frac{c}{3} - \frac{c}{7} = 1$$

$$\frac{4}{21}c = 1 \quad c = \frac{21}{4}$$

(2) 边缘概率密度

$$f_X(x) = \int_{x^2}^1 cxy dy = \frac{21}{8}(x^2 - x^6), -1 \leq x \leq 1$$

$$f_X(x) = \begin{cases} \frac{21}{8}(x^2 - x^6), & -1 \leq x \leq 1 \\ 0, & \text{其他} \end{cases}$$

$$f_Y(y) = \int_{-\sqrt{y}}^{\sqrt{y}} cx^2y dx = cy(\frac{1}{3}x^3) \Big|_{-\sqrt{y}}^{\sqrt{y}} = cy \frac{2}{3}y^{\frac{3}{2}} = \frac{2}{3}cy^{\frac{5}{2}}$$

$$f_Y(y) = \begin{cases} \frac{7}{2}y^{\frac{5}{2}}, & 0 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

(3) 边缘分布律为

$$10. P(X=51) = 0.28 \quad P(X=52) = 0.28 \quad P(X=53) = 0.22 \quad P(X=54) = 0.09$$

$$P(X=55) = 0.13$$

$$P(Y=31) = 0.18 \quad P(Y=32) = 0.15 \quad P(Y=33) = 0.35 \quad P(Y=34) = 0.12$$

$$P(Y=35) = 0.20$$



(2) 条件分布律

$$P(Y=51|X=51) = \frac{P(Y=51, X=51)}{P(X=51)} = \frac{0.06}{0.28} = \frac{6}{28} = \frac{3}{14}$$

$$P(Y=52|X=51) = \frac{P(Y=52, X=51)}{P(X=51)} = \frac{0.07}{0.28} = \frac{1}{4}$$

$$P(Y=53|X=51) = \frac{P(Y=53, X=51)}{P(X=51)} = \frac{0.05}{0.28} = \frac{5}{28}$$

$$P(Y=54|X=51) = \frac{P(Y=54, X=51)}{P(X=51)} = \frac{5}{28}$$

$$P(Y=55|X=51) = \frac{P(Y=55, X=51)}{P(X=51)} = \frac{5}{28}$$

$$f(x, y) = \begin{cases} \frac{21}{4} x^2 y, & x^2 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

13. $f(x, y) = \begin{cases} cx^2 y, & x^2 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$ 在第9题中 $f_X(x) = \frac{21}{8} (x^2 - x^4), -1 \leq x \leq 1$

(1) $f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{\frac{4}{21} x^2 y}{\frac{7}{2} y^{\frac{5}{2}}} = \frac{8}{147} x^2 y^{-\frac{3}{2}}$ $f_Y(y) = \begin{cases} \frac{7}{2} y^{\frac{5}{2}}, & 0 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{\frac{21}{4} x^2 y}{\frac{7}{2} y^{\frac{5}{2}}} = \frac{21}{4} \times \frac{2}{7} x^2 y^{-\frac{3}{2}}$$

$Y = \frac{1}{2}$ 时 X 的条件概率密度

$$f_{X|Y}(x|\frac{1}{2}) = \frac{8}{147} x^2 (\frac{1}{2})^{-\frac{3}{2}} = \frac{8}{147} x^2 \frac{1}{\sqrt{\frac{1}{2}}} = \frac{8}{147} \frac{1}{\frac{1}{\sqrt{2}}} x^2 = \frac{3}{2} x^2 y^{-\frac{3}{2}}$$

$$= \frac{16\sqrt{2}}{147} x^2 \quad Y = \frac{1}{2}$$

(2) $f_{Y|X}(y|x) = \frac{\frac{21}{4} x^2 y}{\frac{21}{8} (x^2 - x^4)} = \frac{21}{4} \times \frac{8}{21} \frac{y}{1-x^4} = \frac{4y}{1-x^4}$

$X = \frac{1}{3}$ $f_{Y|X}(y|x) = \frac{4y}{1-\frac{81}{81}} = \frac{4y}{0} \times \frac{81}{80} y = \frac{81}{40} y \quad \frac{1}{9} \leq y \leq 1$

$X = \frac{1}{2}$ $f_{Y|X}(y|x) = 2 \frac{y}{1-\frac{1}{16}} = 2 \frac{y}{\frac{15}{16}} = \frac{32}{15} y \quad \frac{1}{4} \leq y \leq 1$

$$f_{X|Y}(x|\frac{1}{2}) = \frac{3}{2} x^2 (\frac{1}{2})^{-\frac{3}{2}} = 3 (\frac{1}{2})^{-\frac{1}{2}} x^2 = 3\sqrt{2} x^2$$

$$x^2 = y$$

$$x^2 \leq \frac{1}{2}$$

$$-\frac{\sqrt{2}}{2} \leq x \leq \frac{\sqrt{2}}{2}$$

17. (1) $F(x, y) = \begin{cases} (1-e^{-ax})y, & x \geq 0, 0 \leq y \leq 1, \\ 1-e^{-ax}, & x \geq 0, y > 1, \\ 0, & \text{其他} \end{cases} \quad a > 0$

17 见下页

证明 X, Y 相互独立

由公式, 对于连续型随机变量 $P\{X \leq x, Y \leq y\} = P\{X \leq x\} P\{Y \leq y\}$

即 $F(x, y) = F_X(x) F_Y(y)$ 13. (3) Δ (最后一页会补充)

$$P\{Y \leq \frac{1}{2} | X = \frac{1}{2}\} = \int_{\frac{1}{4}}^{\frac{1}{2}} f_{Y|X}(y|x=\frac{1}{2}) dy = 1$$

$$f(x, y) = \frac{\partial F(x, y)}{\partial x \partial y} = \frac{\partial (1-e^{-ax})}{\partial x} = a e^{-ax}, \quad x \geq 0, 0 \leq y \leq 1$$

$$F_X(x) = \int_{-\infty}^x \int_{-\infty}^{\infty} a e^{-ax} dy dx$$

$$P\{Y \leq \frac{1}{2} | X = \frac{1}{2}\} = \int_{\frac{1}{4}}^{\frac{1}{2}} f_{Y|X}(y|x=\frac{1}{2}) dy = (\frac{16}{15} y^2) \Big|_{\frac{1}{4}}^{\frac{1}{2}} = \frac{16}{15} - \frac{16}{15} \times \frac{1}{16} = \frac{7}{15}$$



17. (1) 设随机变量 (X, Y) 具有分布函数 $F(x, y) = \begin{cases} (1 - e^{-ax})y, & x \geq 0, 0 \leq y \leq 1 \\ 1 - e^{-ax}, & x \geq 0, y > 1 \\ 0, & \text{其他} \end{cases} \quad a > 0.$

证明 X, Y 相互独立

$$P(X \leq x, Y \leq y) = P(X \leq x) P(Y \leq y)$$

$$F(x, y) = F_X(x) F_Y(y)$$

$$F_X(x) = F(x, \infty) = \lim_{y \rightarrow \infty} F(x, y) = \begin{cases} 1 - e^{-ax}, & x \geq 0 \\ 0, & \text{其他} \end{cases}$$

$$F_Y(y) = F(\infty, y) = \lim_{x \rightarrow \infty} F(x, y) = \begin{cases} y, & 0 \leq y \leq 1 \\ 1, & y > 1 \\ 0, & \text{其他} \end{cases}$$

$$\Phi(x) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$$

$$\sqrt{\pi} \Phi(x) = \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$$

$$\int_0^1 e^{-\frac{t^2}{2}} dt = \sqrt{\pi} [\Phi(1) - \Phi(0)]$$

$$= \sqrt{\pi} (0.8413 - 0.5)$$

$$= 0.1405$$

$$\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt = 1$$

$$I = \sqrt{1 - e^{-1}}, \frac{\pi}{2}$$

$$1 - \sqrt{1 - e^{-1}}, \frac{\pi}{2}$$

$$= \int_0^1 (1 - e^{-\frac{x^2}{2}}) dx$$

$$= 1 - \int_0^1 e^{-\frac{x^2}{2}} dx = 1 - \sqrt{\pi} [\Phi(1) - \Phi(0)]$$

$$I = \int_0^1 e^{-\frac{x^2}{2}} dx$$

$$I^2 = \int_0^1 \int_0^1 e^{-\frac{x^2+y^2}{2}} dx dy$$

$$= \int_0^{\frac{\pi}{2}} d\theta \int_0^{\sqrt{2}} e^{-\frac{r^2}{2}} r dr$$

$$= \int_0^{\frac{\pi}{2}} d\theta \int_0^{\sqrt{2}} e^{-\frac{t^2}{2}} d\frac{t^2}{2}$$

$$= \int_0^{\frac{\pi}{2}} d\theta \int_0^1 e^{-t} dt$$

$$= -\int_0^{\frac{\pi}{2}} d\theta \int_0^1 e^t dt$$

$$= (1 - e^{-1}) \frac{\pi}{2}$$

$F(x, y) = F_X(x) F_Y(y)$ 恒成立 故 X, Y 相互独立

18. 由于 X 在区间 $(0, 1)$ 上服从均匀分布

$$\therefore f_X(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{其它} \end{cases} \quad \text{又 } f_Y(y) = \begin{cases} \frac{1}{2} e^{-\frac{y}{2}}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

且 X, Y 独立 故 X, Y 的联合概率密度为

$$f(x, y) = f_X(x) f_Y(y) = \begin{cases} \frac{1}{2} e^{-\frac{y}{2}}, & 0 \leq x \leq 1, y > 0 \\ 0, & \text{其它} \end{cases}$$

(2) a 有实根. 则 $\Delta = 4X^2 - 4Y > 0$ 则 $X^2 - Y > 0$

则 $X^2 > Y$

则 $P(X^2 > Y, 0 \leq x \leq 1, y > 0)$

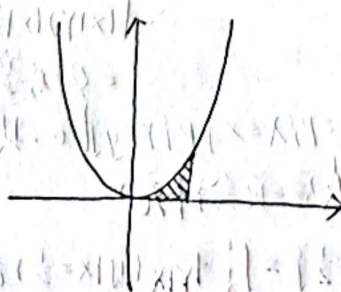
$$= \int_0^1 dx \int_0^{x^2} f(x, y) dy$$

$$= \int_0^1 dx \int_0^{x^2} \frac{1}{2} e^{-\frac{y}{2}} dy$$

$$= \int_0^1 dx \int_0^{x^2} e^{-\frac{t}{2}} d\frac{t}{2}$$

$$= \int_0^1 dx \int_0^{\frac{x^2}{2}} e^{-t} dt$$

$$= -\int_0^1 (e^{-\frac{x^2}{2}} - 1) dx$$



$$22. F_Z = P(Z \leq z) = P\{X+Y \leq z\} = \iint_D f(x,y) dx dy = \int_{-\infty}^{\infty} dy \int_{-\infty}^{z-y} f(x,y) dx$$



$$\text{令 } x = u - y \quad dx = du$$

$$x = z - y \quad u = z \quad F_Z = \int_{-\infty}^{\infty} dy \int_{-\infty}^z f(u-y, y) du \\ = \int_{-\infty}^{\infty} dy \int_{-\infty}^z f(z-y, y) dz \\ = \int_{-\infty}^z dz \int_{-\infty}^{\infty} f(z-y, y) dy$$

$$f_Z = \frac{dF_Z}{dz} = \int_{-\infty}^{\infty} f(z-y, y) dy$$

$$f(z-y, y) = f_X(z-y) f_Y(y) \leftarrow F(x,y) = F_X(x) F_Y(y)$$

$$P\{X \leq x, Y \leq y\} = P\{X \leq x\} P\{Y \leq y\}$$

$$f(x,y) = f_X(x) f_Y(y)$$

由于X, Y独立, 故由上述推导

$$f_Z = \int_{-\infty}^{\infty} f(z-y) f_Y(y) dy$$

$$= \int_{-\infty}^{\infty} e^{-y} dy$$

$$\text{当 } y \in z \in (0, 1) \quad f_Z = \int_0^z e^{-y} dy = -\int_0^z e^{-y} d(-y) = \int_{-z}^0 e^t dt$$

$$z \in (1, \infty) \quad f_Z = \int_{z-1}^z e^{-y} dy = -\int_{z-1}^z e^{-y} d(-y) = \int_{-z}^{-z+1} e^t dt = 1 - e^{-z}$$

其它 $f_Z = 0$.

$$24. f(x,y) = \begin{cases} \frac{1}{2}(x+y)e^{-(x+y)}, & x > 0, y > 0 \\ 0, & \text{其他} \end{cases}$$

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y) dy$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y) dx$$

$$= \int_{-\infty}^{+\infty} f \left(\frac{1}{2}(x+y) \right) e^{-(x+y)} dy = f(x+y)$$

$$= \int_{-\infty}^{+\infty} \frac{1}{2} (x+y) e^{-(x+y)} d(x+y) \quad \text{用分部积分法 反对幂三指}$$

$$= \int_{-\infty}^{+\infty} \frac{1}{2} t e^{-t} dt$$

$$= \int_0^{\infty} -\frac{1}{2} (x+y) d e^{-(x+y)}$$

$$t = (x+y) \cdot \int_0^{\infty} -\frac{1}{2} (x+y) e^{-(x+y)} d(x+y) = \int_{-\infty}^{\infty} -\frac{1}{2} t e^t dt$$

$$\int_{-\infty}^{-x} e^t dt = [e^t]_{-\infty}^{-x}$$

$$= \frac{1}{2} (t+1) \left\{ [te^t]_{-x}^{\infty} - \int_{-x}^{\infty} e^t dt \right\}$$

$$= \frac{1}{2} (x e^{-x} + e^{-x})$$

$$= \frac{1}{2} (1+x) e^{-x}$$

$$\text{同理 } f_Y(y) = \frac{1}{2} (1+y) e^{-y}$$

$$\text{显然 } f(x,y) = \frac{1}{2} (x+y) e^{-(x+y)} \neq f_X(x) f_Y(y)$$

故X和Y不是相互独立的



(2) $Z = X + Y$ 的概率密度

$X > 0, Y > 0 \therefore Z > 0$

$$f_Z(z) = F_Z'(z) = \int_{-\infty}^{\infty} f(z-y, y) dy = \int_{-\infty}^{\infty} \frac{1}{2} (z-y+y) e^{-(z-y+y)} dy$$

$z-y > 0$
 $z > y$ ✓

$$= \int_{-\infty}^{\infty} \frac{1}{2} z e^{-z} dy$$

$$= \int_0^z \frac{1}{2} z e^{-z} dz$$

$$= -\frac{1}{2} \int_0^z z e^{-z} d(z)$$

$$= -\frac{1}{2} \int_0^z (-z) e^{-z} d(-z) = \frac{1}{2} \int_0^z t e^t dt$$

$$t = -z \quad z = -z$$

$$= \frac{1}{2} \int_0^z t e^t dt = \frac{1}{2} \left[t e^t \right]_0^z - \int_0^z e^t dt$$

$$= \frac{1}{2} (z e^{-z} - e^{-z} + 1)$$

抄错了

$$\therefore f_Z = \begin{cases} \frac{1}{2} (1 - z e^{-z} - e^{-z}), & z > 0 \\ 0, & z \leq 0 \end{cases}$$

设

$$\frac{1}{2} \int_0^z z e^{-z} dy = [z e^{-z} y]_0^z = \frac{1}{2} z^2 e^{-z}, \quad z > 0$$

$$\therefore (1) F(x, y) = \int_0^1 dx \int_0^{\infty} b e^{-x+y} dy$$

$$= b \int_0^1 e^{-x} dx \int_0^{\infty} e^{-y} dy$$

$$= b \int_0^1 e^{-x} d(-x) \int_0^{\infty} e^{-y} d(-y)$$

$$= b \int_0^1 e^t dt \int_0^{\infty} e^t dt$$

$$= b (e^{-1} - 1) (-1) = (1 - \frac{1}{e}) b = \frac{e-1}{e} b = 1$$

$$b = \frac{e}{e-1}$$

(2) 求边缘概率密度 $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$

以后先写好范围

$$= b e^{-x} \int_{-\infty}^{\infty} e^{-y} dy = -b e^{-x} \int_{-\infty}^{\infty} e^{-y} d(-y) = -b e^{-x} [e^{-y}]_{-\infty}^{\infty}$$

$$= -b e^{-x} \int_{-\infty}^{\infty} e^t dt$$

$$= b e^{-x} \int_{-\infty}^{\infty} e^t dt = b e^{-x}$$

$$= \frac{e}{e-1} e^{-x} = \frac{e^{-x}}{e-1}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$= \int_0^1 f(x, y) dx$$

$$= b e^{-y} \int_0^1 e^{-x} dx$$

$$= b e^{-y} [-e^{-x}]_0^1 = b e^{-y} (-e^{-1} + 1) = \frac{e}{e-1} e^{-y} \frac{e-1}{e} = e^{-y}$$

(3) $U_{\max}\{X, Y\}$ 的分布函数 $F_M(z) = P(M \leq z) = P(X \leq z) P(Y \leq z) = F_X(z) F_Y(z)$

$z \in (0, 1)$ 时

$$F_X(z) = \int_{-\infty}^z f_X(x) dx = \frac{e}{e-1} \int_{-\infty}^z e^{-x} dx = (-e^{-x})_0^z$$

$z \geq 1$ 时

$$F_X(z) = \frac{e}{e-1} \int_0^1 f_X(x) dx = (-e^{-x})_0^1 \cdot \frac{e}{e-1} = \frac{e}{e-1} (1 - e^{-1}) = \frac{e - e^{-1}}{e-1}$$

$$F_X(z) = \begin{cases} \frac{e - e^{-z}}{e-1}, & z \in (0, 1) \\ 1, & z \geq 1 \\ 0, & \text{其他} \end{cases}$$

$$F_Y(z) = \int_{-\infty}^z f_Y(y) dy = \int_0^z e^{-y} dy = (-e^{-y})_0^z = 1 - e^{-z}$$

$$F_Y(z) = \begin{cases} 1 - e^{-z}, & z > 0 \\ 0, & z \leq 0 \end{cases}$$

$$\therefore F_M(z) = \begin{cases} \frac{e}{e-1} (1 - e^{-z})^2, & z \in (0, 1) \\ 1 - e^{-z}, & z \geq 1 \\ 0, & z \leq 0 \end{cases}$$



设有 Y 只寿命小于 180 根则

30 $Y \sim B(4, p)$ 而某种型号电子元件的寿命为 X , $X \sim N(160, 20^2)$

则 $p = P(X \geq 180)$

$$= 1 - P(X \leq 180) = 1 - P(X \leq \mu + \sigma) = 1 - \left(\frac{68.26\%}{2} + \frac{1}{2} \right)$$

$$= 1 - (0.3413 + 0.5)$$

$$\text{没有一只寿命小于 180 的概率为 } p = (0.1587)^4 = 0.00063$$

